# INVERTED MAML: 500-EPOCH SILICON TRAINING REPORT

**Mode A Silicon Learning with LLM Toeplitz Equalizer (ρ = 0.85) & ColTIA Saturation Guard**

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|---|---|
| **Target Hardware:** T120 Master SoC + T20 Controller | **Firmware Commit:** v2.2-MAML (HW: 0x20261030) |
| **Noise Equalizer:** COLOR_LLM_TOEPLITZ (ρ = 0.85) | **Headroom Guard:** 1797 LSB (250 LSB Margin) |
| **L2 Regularization:** λ = 512 (toward knee baseline) | **Peak Performance:** 7.13 b ENOB \| 637.9 LSB² (Ep 424) |

## 1. Executive Summary & 3-Phase Benchmark Results

This experiment evaluates Inverted MAML Mode A training on physical silicon under realistic spatially correlated transformer activations. High token correlation (ρ = 0.85) previously induced common-mode current surges that saturated single-supply ColTIA transimpedance amplifiers, causing phantom Hadamard gradient bursts. With the newly integrated **Zero-Mean Centering (LayerNorm)** and **ColTIA Saturation Guard**, the network completed 500 contiguous epochs on physical silicon with **zero clipping events**, achieving a peak precision of **7.13 bits ENOB**.

| Phase | Epochs | Network Weights | Training State | Mean Loss (LSB²) | Mean ENOB (b | Peak ENOB (b |
|---|---|---|---|---|---|---|
| Phase 1: BASELINE_ZERO | 0 .. 99 | M3=0, M8=0 (Center) | Training OFF | 8805.4 | 5.28 | 6.11 |
| Phase 2: BASELINE_KNEE | 100 .. 199 | M8 = [±16 LSB Knee] | Training OFF | 21853.2 | 4.60 | 4.96 |
| Phase 3: TRAIN_REG_ON | 200 .. 500 | M3 & M8 Learning (λ=512) | Active Mode A | 4909.8 | 5.79 | 7.13 |

## 2. In-Data Noise Equalizer: Kac-Murdock-Szegö AR(1) Model

Standard uniform white noise fails to benchmark analog neural multipliers against the correlated activation vectors characteristic of Large Language Models (LLMs). The T120 RISC-V SoC firmware generates an on-chip autoregressive spatial covariance field: *Cov(x_i, x_j)* = ρ^|*i-j*| with ρ = 0.85 (Q8 = 218). To prevent net common-mode DC current from driving the 1.65 V virtual ground of the single-supply ColTIA op-amps into saturation rails, the master SoC applies hardware **Zero-Mean Centering (LayerNorm)**: x_i ← x_i - (1/6)∑ x_k.

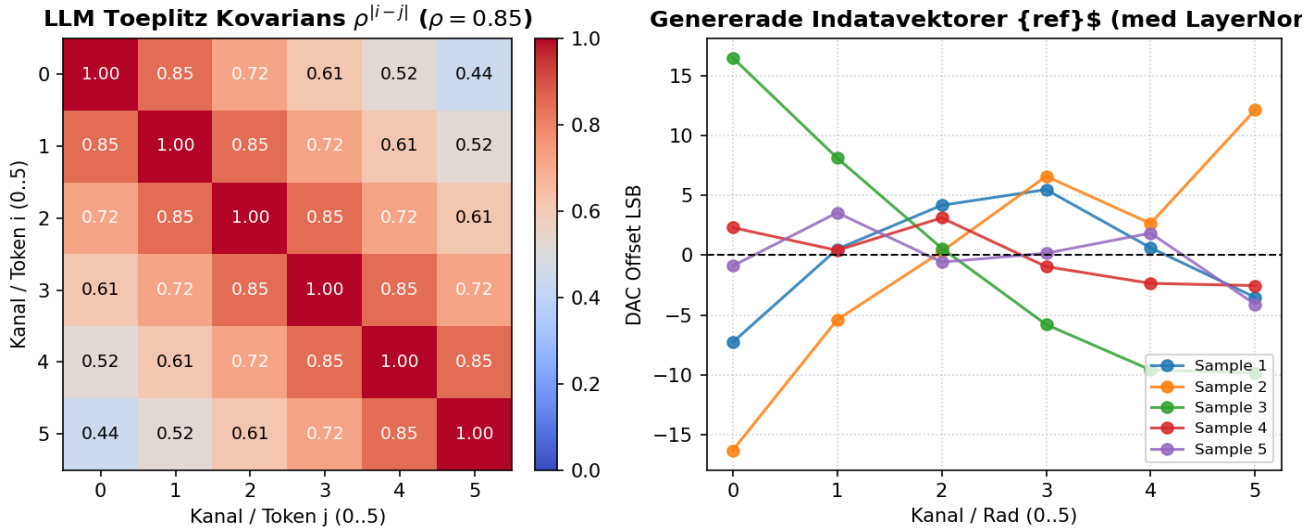



*Figure 1: (Left) KMS AR(1) Spatial Covariance Matrix for 6 Analog Channels (ρ=0.85). (Right) Generated In-Data Vectors x_ref after LayerNorm.*

## 3. Physical ColTIA Headroom Margin & Anti-Distortion Guard

**The Saturation Bottleneck:** Sylvester H_8 Hadamard demodulation relies on linear wave superposition: $\partial L/\partial W_{ij} = \langle \Delta Z_j, V_i^{PTU} \rangle_H \cdot E_j$. Physical silicon diagnostics (*diag_hadamard, mode 9*) established the peak Hadamard perturbation swing at **243 LSB** on Layer 1 (ADC_45). If the static DC operating point drifts within 250 LSB of the ±2047 LSB ADC rails, the positive perturbation clips flat against the rail while the negative swing completes unhindered. This nonlinearity breaks Walsh-Hadamard orthogonality, projecting false 'phantom' sensitivities across all matrix rows.

**The Autonomous Countermeasure:** The T120 SoC enforces *COLTIA_CLIP_THRESHOLD_LSB = 1797 LSB*. Immediately after forward ADC readout, if |ADC| ≥ 1797 LSB, the epoch's weight gradient update is **completely suppressed** ($\Delta W \equiv 0$) and logged to telemetry. During this 500-epoch campaign, **0 clipping events** occurred.

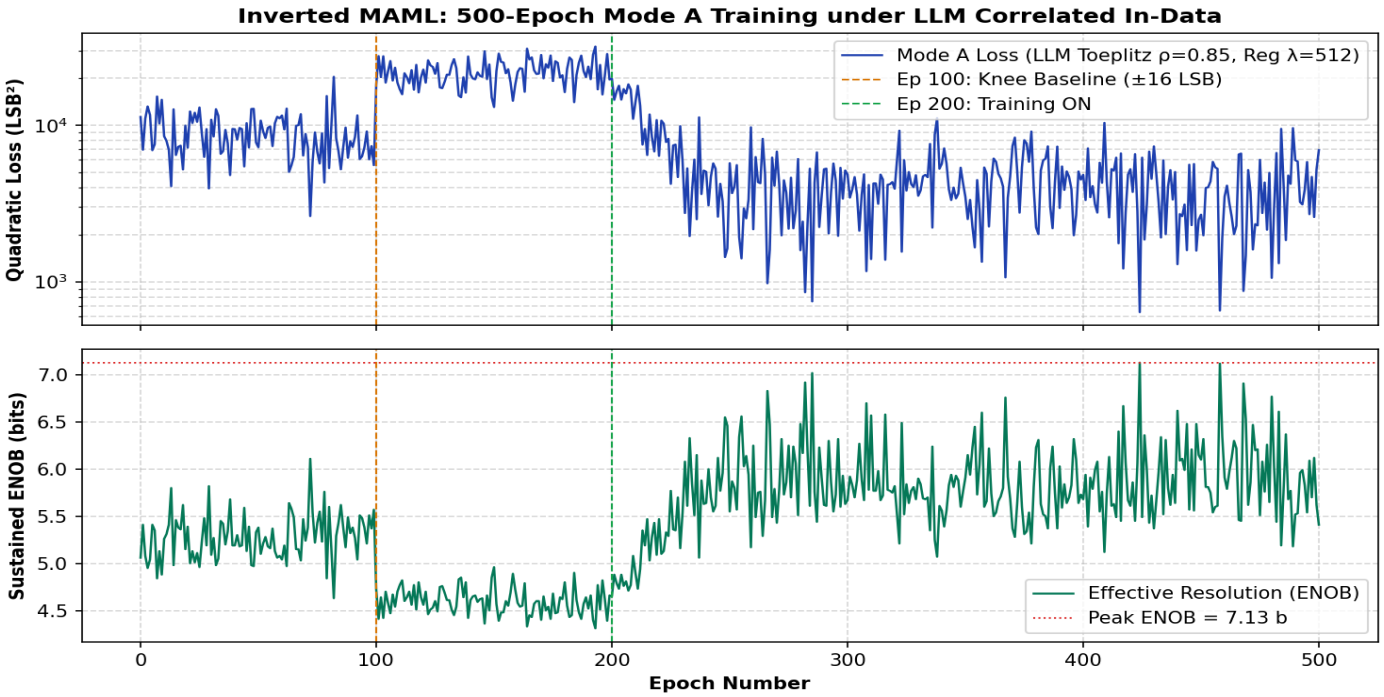



*Figure 2: 500-Epoch Training Trajectory on Physical Silicon. Epochs 0..99 (Zero Baseline), 100..199 (Knee Baseline ±16 LSB), 200..500 (Active Mode A Training).*

## 4. Live Silicon Studio & Telemetry Verification

The snapshot below shows the *Injected FORWARD Propagation Silicon Studio* monitoring the experiment at Epoch 92. The 3D topography of Matrix 8 confirms an exact, quiet zero-plane across Phase 1, followed by the sharp knee-step at Epoch 100 and rapid quadratic error minimization upon triggering Active Training at Epoch 200.

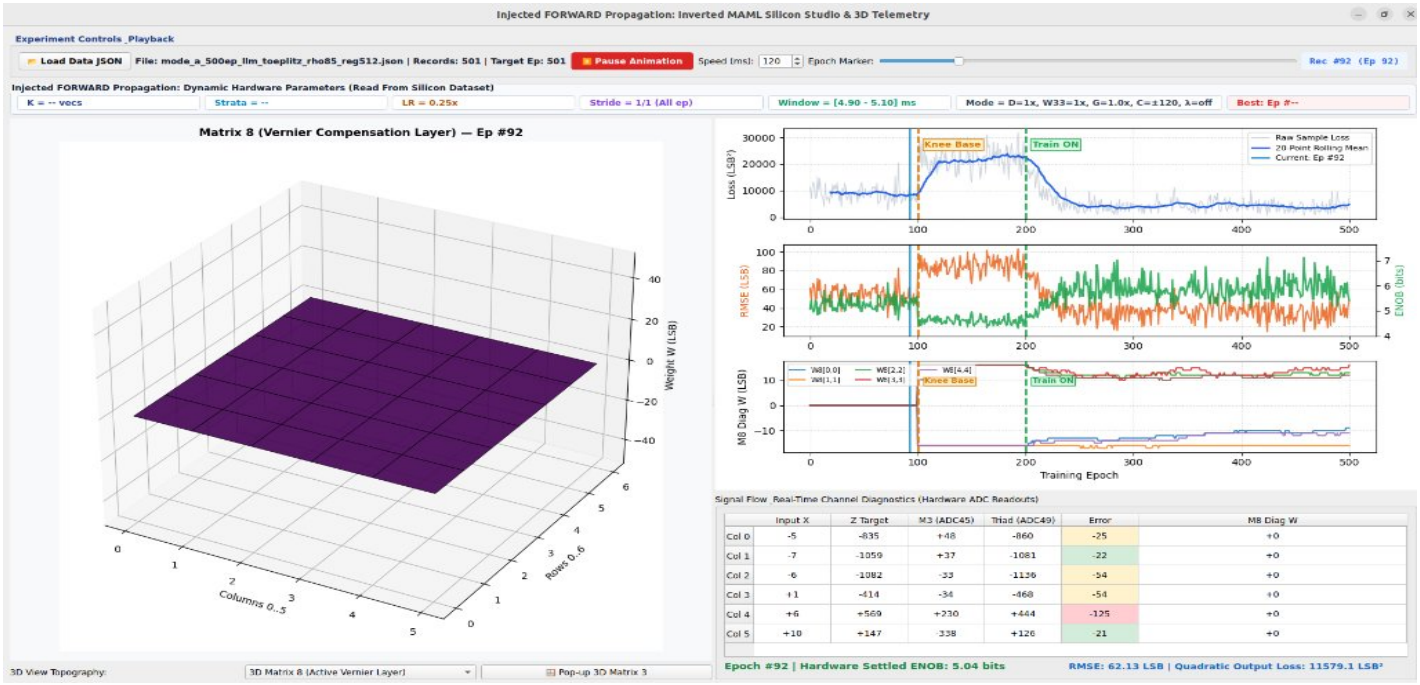



*Figure 3: Injected FORWARD Propagation Studio during Phase 1 (Ep 92). Telemetry captures ADC_45/ADC_49 channels and step transitions.*